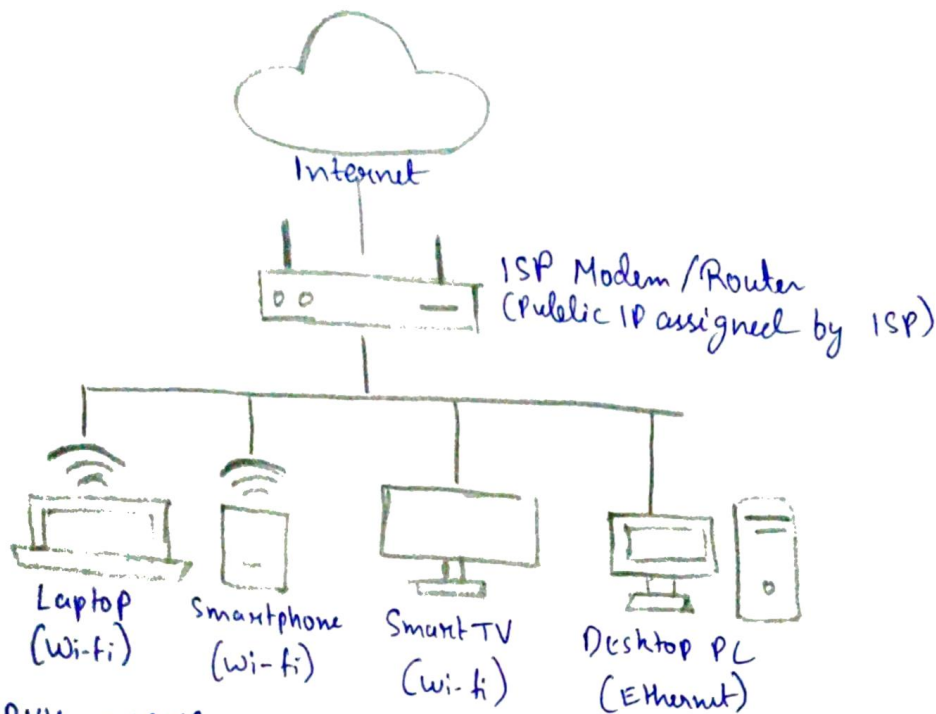


Assignment : 1

Home Network Topology -



i) IPv4 vs IPv6

- IPv4 uses 32-bit addresses, allowing ~4.3 billion addresses
- IPv6 uses 128-bit addresses, allowing trillions of devices
- IPv6 solves the address exhaustion problem and offers better security and routing efficiency.

ii) Subnetting basics:-

Subnetting divides a large network into smaller sub-networks (subnets). It helps manage IP allocation, improves security, and reduces congestion.

iii) Network address Translation (NAT):-

NAT allows multiple devices in a private network to access the internet using a single public IP address. It maps private IPs to the public IP at the router level, conserving IPv4 addresses and adding a layer of security.

iv) Advanced IP address management strategies:-

These include using DHCP with reservation, IPAM (IP Address Management) tools, subnet planning & IP usage monitoring. They help in organizing large-scale networks & avoiding IP conflicts.

v) IP address allocation & policies:-

IP addresses are allocated dynamically (via DHCP) or statically. Allocation follows standards set by IANA & RIRs. Policies ensure efficient usage, like avoiding conflicts, assigning based on subnetting & respecting reserved address ranges (RFC 1918).

vii) Autonomous System (AS) numbers & their importance:-

An (AS) is a network or group of networks under one organization with a unified routing policy. Each AS is assigned a unique number (ASN). ASNs are crucial for BGP routing between large networks on the internet.

viii) Global IP addressing issues & solutions:-

IPv4 address exhaustion occurred due to the limited number of 32-bit addresses. Solutions include:

- IPv6 adoption (128-bit addressing),
- NAT usage to share IPs,
- Better IP management
- Encouraging address reuse & efficient subnetting.

ix) Comparison of wired & wireless networking.

Feature	Wired	Wireless
Speed	Generally faster	Slower than wired
Reliability	High	Susceptible to drops
Mobility	Limited	Excellent
Setup	More complex	Easier

x) WiFi, Ethernet & emerging technologies:-

- WiFi: Wireless networking using radio waves.
- Ethernet: wired LAN connection, stable & fast
- Emerging: Wi-Fi 7, 5G Integration & wireless mesh networking enhance coverage & speed.

xi) Powerline & optical fiber communication:-

- Powerline: Uses electrical wiring to transmit data; useful where Wi-Fi is weak.
- Optical fibre: Uses light signals through glass cables; offers ultra-high speed & long-distance reliability.

xii) Advanced wireless technologies:-

LiFi uses LED light to transmit data at high speeds. It is secure, fast & doesn't interfere with RF signals - ideal for sensitive environments like hospitals.

xiii) Performance Analysis & comparison of network types:-

- wired: offers consistent high-speed & low latency - ideal for gaming, streaming, & work
- wireless: offers flexibility & mobility but may suffer in speed & signal reliability.
- Performance depends on distance, interference, bandwidth & device capability.